

A Twelve-Value Proof of the A181280 Formula

Fourier spectra and universal recurrences for ordered binary Gram matrices

ChatGPT

AI-assisted mathematical research report

19 September 2026

Abstract

OEIS A181280 counts binary matrices with four lexicographically increasing rows whose mod-2 Gram matrix has lexicographically decreasing rows. Kauers and Koutschan proposed an explicit six-term exponential-polynomial formula in 2023. We give a computer-assisted proof of that formula: a structural argument first proves a universal order-12 recurrence for the tail, after which twelve exact counts settle the identity for every $n \geq 4$. The finite arithmetic is supplied in two independent implementations, with a further Fourier audit and direct brute-force checks.

The structural argument works for any fixed number m of rows and any set of allowed symmetric Gram matrices. It gives an explicit eventual recurrence of order $m + 2\lfloor m^2/4 \rfloor$, with characteristic roots restricted to signed powers of two and explicitly bounded multiplicities. For four rows this universal order is sharp. We also determine the rational generating function, the minimal eventual recurrence of order 11 for A181280, the exact transient, and quantitative equidistribution and asymptotic consequences.

Status and attribution

The OEIS formula was still labeled conjectural when retrieved on 19 September 2026 [1]. However, Kauers's December 2025 slides already prove the general *existence* of a constant-coefficient recurrence [3]. That prior result is credited here, not claimed as new. The slides display the explicit formula with a question mark but also mark A181280 with a check in an overview table; they do not supply an explicit formula certificate. Thus the exact prior-resolution status is not completely settled by the available sources. This report provides a self-contained, reproducible proof, but makes no claim of priority and has not received independent peer review.

Keywords. OEIS A181280; finite-state enumeration; binary quadratic forms; Walsh sums; C-finite sequences; rational generating functions.

Contents

1	The selected problem and the result	3
1.1	The counting problem	3
1.2	What was already known, and what is established here	3
1.3	The key reduction	3
2	A prefix-order automaton	4
2.1	General row count and the Gram statistic	4
2.2	Remembering the first differences	4
2.3	An exact integer dynamic program	5
3	Fourier decomposition and quadratic Walsh sums	5
3.1	Characters of the symmetric-matrix group	5
3.2	Small weighted transition matrices	6
3.3	A self-contained Walsh-sum lemma	6
3.4	Why the diagonal entries are Walsh sums	7
4	An explicit universal recurrence	7
4.1	An annihilator from the level filtration	7
4.2	The four-row specialization	9
5	The finite certificate and proof of the formula	9
5.1	Encoding the finite calculation	9
5.2	Why the twelve checks finish the proof	10
6	Generating functions, minimal recurrences, and sharpness	11
6.1	The exact rational generating function	11
6.2	Minimality for the selected sequence	12
6.3	The four-row universal bound is sharp	12
7	Equidistribution and exact asymptotic information	13
7.1	A general uniformity theorem	13
7.2	Leading terms for A181280	13
7.3	All coefficients of the conditional-probability expansion	14
8	Computation, reproducibility, and audit boundaries	14
8.1	What was executed	14
8.2	What makes this a finite proof rather than a large test	15
8.3	Complexity and practical use	15
9	Conclusions and remaining research directions	15
A	The recurrence certificate in machine-readable form	16
B	A complete small arithmetic verifier	17
C	Package contents and reproduction	18

1 The selected problem and the result

1.1 The counting problem

For a binary word, lexicographic order uses $0 < 1$ and reads from left to right. All words being compared have the same length. Let $a(n)$ be the number of matrices $M \in \mathbb{F}_2^{4 \times n}$ such that

$$M_{1,*} <_{\text{lex}} M_{2,*} <_{\text{lex}} M_{3,*} <_{\text{lex}} M_{4,*}, \quad (MM^T)_{1,*} >_{\text{lex}} (MM^T)_{2,*} >_{\text{lex}} (MM^T)_{3,*} >_{\text{lex}} (MM^T)_{4,*}. \quad (1)$$

The product is taken over $\mathbb{F}_2$, not over the integers. Thus its (i, j) entry is the parity of the ordinary dot product of rows i and j . For $n \geq 1$ this is the sequence A181280, introduced in the OEIS by R. H. Hardin [1]. We extend it by $a(0) = 0$: four empty rows are not strictly ordered.

Kauers and Koutschan state the proposed formula as Conjecture 20 in Section 6.5 of their 2023 paper [2]. The following is that same formula, rewritten to group equal exponential bases.

Theorem 1.1 (The A181280 formula). *For every integer $n \geq 4$,*

$$\begin{aligned} a(n) = & \frac{16^n}{512} + \frac{288n - 3473}{147456} 8^n - \frac{113}{49152} (-8)^n \\ & + \frac{6n^2 - 219n + 820}{6144} 4^n - \frac{13n - 164}{6144} (-4)^n - \frac{3n + 32}{288} 2^n. \end{aligned} \quad (2)$$

Also $a(0) = a(1) = a(2) = a(3) = 0$.

The proof occupies Sections 2–5. The only computer-assisted part essential to the theorem is an exact finite enumeration through $n = 15$. The reason this is sufficient is proved before the counts are used. No guessed recurrence is assumed in that proof.

1.2 What was already known, and what is established here

Kauers's talk *Guessing with little data*, listed in the Rutgers seminar archive for 11 December 2025 [4], contains a transfer-matrix and parity-filtering proof that these counting sequences are C-finite for arbitrary fixed row count and arbitrary allowed Gram set [3]. Here *C-finite* means satisfying a linear recurrence with constant coefficients. Its formula slide is numbered 30, and its general C-finiteness theorem is on slide 39. Because the PDF contains overlay pages, these are PDF pages 104 and 137, respectively, counting pages from one.

The contribution developed in this report is more explicit than existence: we locate the possible characteristic roots, bound their multiplicities using a filtration with only m levels, prove a quadratic-in- m universal recurrence order, and use that bound to finish a finite certificate for the displayed formula. We then prove sharpness of the four-row universal bound and derive several exact consequences. The literature check located the 2023 conjecture, the live OEIS entry, and the 2025 slides, but not a separate complete proof of the explicit formula. That search is not a proof that no such proof exists.

1.3 The key reduction

A direct automaton would have $8 \cdot 1024 = 8192$ states: eight row-order states and 1024 possible symmetric 4×4 Gram matrices. The key fact is that its counting sequences have a much smaller universal annihilator.

With E denoting the forward shift, $(Ef)(n) = f(n+1)$, define

$$P_4(x) = (x-2)^2(x+2)(x-4)^3(x+4)^2 \cdot (x-8)^2(x+8)(x-16). \quad (3)$$

We prove

$$P_4(E)a(n) = 0 \quad (n \geq 4). \quad (4)$$

This monic polynomial has degree 12. The right side of (2) satisfies the same recurrence. Therefore equality at $n = 4, 5, \dots, 15$ proves equality at every later index by induction. Establishing (4) from the definition, rather than inferring it from initial data, is what turns the calculation into a proof.

2 A prefix-order automaton

2.1 General row count and the Gram statistic

Fix $m \geq 1$ and put

$$d = \frac{m(m+1)}{2}, \quad \mathcal{G}_m = \text{Sym}_m(\mathbb{F}_2).$$

For $\mathcal{A} \subseteq \mathcal{G}_m$, let $N_{m,\mathcal{A}}(n)$ count matrices $M \in \mathbb{F}_2^{m \times n}$ with strictly increasing rows and $MM^\top \in \mathcal{A}$. The special set for our problem is

$$\mathcal{A}_* = \{G \in \mathcal{G}_4 : G_{1,*} >_{\text{lex}} G_{2,*} >_{\text{lex}} G_{3,*} >_{\text{lex}} G_{4,*}\}, \quad a(n) = N_{4,\mathcal{A}_*}(n).$$

Allowing only symmetric targets loses nothing, since every Gram matrix here is symmetric. If an allowed set initially contains nonsymmetric matrices, they can simply be discarded.

Write a matrix as a sequence of columns $v_1, \dots, v_n \in \mathbb{F}_2^m$. Its Gram matrix is

$$MM^\top = \sum_{t=1}^n v_t v_t^\top. \quad (5)$$

Thus appending a column v adds $vv^\top$ to the current Gram matrix.

2.2 Remembering the first differences

After reading a prefix of the columns, record the subset

$$S \subseteq \{1, \dots, m-1\}$$

of adjacent row pairs whose first difference has already occurred in the correct direction. On every retained prefix, $i \in S$ means that row i is already lexicographically smaller than row $i+1$; $i \notin S$ means their prefixes are still equal. Once resolved, a pair never needs to be compared again.

A new column $v = (v_1, \dots, v_m)$ is admissible from S exactly when

$$v_i \leq v_{i+1} \quad \text{for every } i \notin S. \quad (6)$$

For an admissible column, the new state is

$$\tau(S, v) = S \cup \{i : v_i = 0, v_{i+1} = 1\}. \quad (7)$$

The initial state is $\emptyset$ and the accepting order state is $S_{\text{full}} = \{1, \dots, m-1\}$. There are 2^{m-1} order states.

Lemma 2.1 (Correctness of the order automaton). *Length- n paths from $\emptyset$ to S_{full} , with columns as transition labels and with admissibility (6), are in bijection with binary $m \times n$ matrices having strictly increasing rows.*

Proof. At the empty prefix all adjacent pairs are equal. Suppose the description of a state is correct for a retained prefix. For an unresolved pair, a new pair of bits $(1, 0)$ makes the first difference wrong and must be rejected; $(0, 0)$ or $(1, 1)$ preserves equality; $(0, 1)$ resolves it correctly. For a resolved pair, any later bits leave its first difference unchanged. This proves the invariant and the transition rule by induction.

At the end, every adjacent pair must be resolved to obtain strict order. Conversely, any strictly increasing matrix has a first difference of type $(0, 1)$ for each adjacent pair and therefore follows exactly one accepted path. Adjacent strict inequalities are equivalent to the whole increasing row chain. $\square$

2.3 An exact integer dynamic program

Let $D_n(S, G)$ count retained prefixes of length n with order state S and Gram matrix G . Start with

$$D_0(\emptyset, 0) = 1, \quad D_0(S, G) = 0 \quad \text{otherwise.} \quad (8)$$

For each admissible v , perform the update

$$D_{n+1}(\tau(S, v), G + vv^T) += D_n(S, G). \quad (9)$$

Then

$$N_{m, \mathcal{A}}(n) = \sum_{G \in \mathcal{A}} D_n(S_{\text{full}}, G). \quad (10)$$

These equations, together with Theorem 2.1 and (5), prove correctness of the finite enumeration used later. Different columns that lead to the same next state must be counted separately; the update does so.

An important independent total is

$$\sum_{G \in \mathcal{G}_m} D_n(S_{\text{full}}, G) = \binom{2^n}{m}. \quad (11)$$

Indeed, a strictly increasing list of m binary words of length n is just an m -element subset of the 2^n possible words, written in its unique increasing order. The supplied implementation checks this equality at every computed length.

3 Fourier decomposition and quadratic Walsh sums

3.1 Characters of the symmetric-matrix group

Identify $\mathcal{G}_m$ with $\mathbb{F}_2^d$ using its upper-triangular entries. For $Q = (Q_{ij})_{i \leq j} \in \mathbb{F}_2^d$ and $G \in \mathcal{G}_m$, define

$$\langle Q, G \rangle = \sum_{i \leq j} Q_{ij} G_{ij} \pmod{2}, \quad \chi_Q(G) = (-1)^{\langle Q, G \rangle}. \quad (12)$$

This is an upper-triangle pairing, *not* the ordinary trace pairing of two symmetric matrices in characteristic two. Using a trace pairing would count off-diagonal entries twice and destroy the required character group.

Let

$$\widehat{\mathbf{1}}_{\mathcal{A}}(Q) = \sum_{G \in \mathcal{A}} \chi_Q(G).$$

The elementary orthogonality identity

$$\sum_{Q \in \mathbb{F}_2^d} (-1)^{\langle Q, H \rangle} = \begin{cases} 2^d, & H = 0, \\ 0, & H \neq 0 \end{cases}$$

implies

$$\mathbf{1}_{\mathcal{A}}(G) = 2^{-d} \sum_{Q \in \mathbb{F}_2^d} \widehat{\mathbf{1}_{\mathcal{A}}}(Q) \chi_Q(G). \quad (13)$$

For completeness, when $H \neq 0$, choose a coordinate where it is 1 and pair each Q with the vector obtained by toggling that coordinate; the two signs cancel. Applying this to $H = G + G'$ proves the inversion formula.

3.2 Small weighted transition matrices

For $Q \in \mathbb{F}_2^d$, introduce the quadratic Boolean function

$$q_Q(v) = \langle Q, vv^\top \rangle = \sum_i Q_{ii} v_i + \sum_{i < j} Q_{ij} v_i v_j. \quad (14)$$

The diagonal terms are linear because $v_i^2 = v_i$ on $\mathbb{F}_2$. Define the $2^{m-1} \times 2^{m-1}$ integer matrix T_Q by

$$(T_Q)_{S, S'} = \sum_{\substack{v \text{ admissible from } S \\ \tau(S, v) = S'}} (-1)^{q_Q(v)}. \quad (15)$$

Vectors $\mathbf{e}_S$ are standard coordinate columns. The weighted number of accepted paths is

$$b_Q(n) = \mathbf{e}_\emptyset^\top T_Q^n \mathbf{e}_{S_{\text{full}}}.$$

Since χ_Q turns the Gram sum into a product of column weights, (13) gives

$$N_{m, \mathcal{A}}(n) = 2^{-d} \sum_{Q \in \mathbb{F}_2^d} \widehat{\mathbf{1}_{\mathcal{A}}}(Q) \mathbf{e}_\emptyset^\top T_Q^n \mathbf{e}_{S_{\text{full}}}. \quad (16)$$

For four rows, this replaces one 8192-state system by 1024 integer matrices of size eight. More importantly, all these small matrices have tightly controlled spectra and the same short filtration.

3.3 A self-contained Walsh-sum lemma

Lemma 3.1 (Possible quadratic sums). *Let $q : \mathbb{F}_2^k \rightarrow \mathbb{F}_2$ be a polynomial function of degree at most two with $q(0) = 0$, and put*

$$W(q) = \sum_{v \in \mathbb{F}_2^k} (-1)^{q(v)}.$$

Then

$$W(q) \in \Lambda_k := \{0, 2^k\} \cup \{\pm 2^{k-s} : 1 \leq s \leq \lfloor k/2 \rfloor\}. \quad (17)$$

Proof. The polarization

$$B(u, v) = q(u + v) + q(u) + q(v)$$

is an alternating bilinear form. Its rank r is even. One elementary proof is by induction: if B is nonzero, choose u, v with $B(u, v) = 1$; their span is a nondegenerate two-dimensional summand, and its orthogonal complement reduces the rank by two. If $B = 0$, its rank is zero.

Let $R = \text{rad } B$, so $\dim R = k - r$. Squaring the sum and substituting $w = v + u$ gives

$$\begin{aligned} W(q)^2 &= \sum_{u, v \in \mathbb{F}_2^k} (-1)^{q(v) + q(v+u)} \\ &= \sum_{u \in \mathbb{F}_2^k} (-1)^{q(u)} \sum_{v \in \mathbb{F}_2^k} (-1)^{B(v, u)} = 2^k \sum_{u \in R} (-1)^{q(u)}. \end{aligned}$$

On R the function q is linear, since its polarization vanishes there. The last sum is either zero or 2^{k-r} . Hence $W(q) = 0$ or $W(q) = \pm 2^{k-r/2}$.

There is one sign restriction to retain. If $r = 0$, then q is linear on all of $\mathbb{F}_2^k$. Its sum is zero when it is nonzero and 2^k when it is zero; it cannot be -2^k . For positive even $r = 2s$, the remaining possibilities are exactly those permitted by (17). The argument only needs this inclusion, not that every listed value occurs in every application. $\square$

3.4 Why the diagonal entries are Walsh sums

A transition can stay at the same order state S only when

$$v_i = v_{i+1} \quad (i \notin S).$$

These equations say that v is constant on each contiguous block separated by the boundaries in S . There are $k = |S| + 1$ such blocks. Assigning a bit to each block parametrizes the allowed staying columns by $\mathbb{F}_2^k$. Restricting q_Q to this subspace preserves its degree bound and its zero constant term. Therefore

$$(T_Q)_{S,S} \in \Lambda_{|S|+1}. \quad (18)$$

Also a nonzero transition only moves from S to a superset S' . In particular, it never moves to a different state on the same cardinality level. This second fact controls multiplicities; the diagonal values alone would not suffice.

4 An explicit universal recurrence

4.1 An annihilator from the level filtration

Lemma 4.1 (Filtered annihilation). *Suppose T preserves a filtration $0 = V_0 \subseteq V_1 \subseteq \dots \subseteq V_m = V$. If the induced operator on V_k/V_{k-1} is annihilated by a polynomial p_k , then $\prod_{k=1}^m p_k(T) = 0$.*

Proof. The assumption is exactly $p_k(T)V_k \subseteq V_{k-1}$. Apply $p_m(T)$ first, then $p_{m-1}(T)$, down to $p_1(T)$. Their product maps V_m into $V_0 = 0$. All factors are polynomials in the same operator, so they commute; their written order is immaterial. $\square$

Order the coordinate basis by $|S|$. Regard T_Q as acting on column vectors, and let V_k be the span of $\mathbf{e}_S$ with $|S| \leq k - 1$. Because $(T_Q)_{S,S'}$ can be nonzero only when $S \subseteq S'$, these subspaces are invariant. On the

quotient V_k/V_{k-1} the matrix is diagonal. Thus (18) and Theorem 4.1 apply with

$$p_k(x) = \prod_{\lambda \in \Lambda_k} (x - \lambda) = x(x - 2^k) \prod_{s=1}^{\lfloor k/2 \rfloor} (x - 2^{k-s})(x + 2^{k-s}). \quad (19)$$

The polynomial has only one copy of each allowed diagonal value at that level. We do not multiply a characteristic polynomial for every state.

Theorem 4.2 (Universal spectral recurrence). *For $m \geq 1$, set $r_j = \min(j, m - j)$ and define*

$$P_m(x) = \prod_{j=1}^m (x - 2^j)^{r_j+1} (x + 2^j)^{r_j}, \quad (20)$$

$$U_m(x) = x^m P_m(x). \quad (21)$$

For every $\mathcal{A} \subseteq \mathcal{G}_m$,

$$U_m(E)N_{m,\mathcal{A}}(n) = 0 \quad (n \geq 0), \quad P_m(E)N_{m,\mathcal{A}}(n) = 0 \quad (n \geq m). \quad (22)$$

Moreover,

$$\deg P_m = m + 2\lfloor m^2/4 \rfloor, \quad \deg U_m = 2m + 2\lfloor m^2/4 \rfloor. \quad (23)$$

Proof. Equations (19) and the filtration lemma show that $\prod_{k=1}^m p_k(T_Q) = 0$ for every Q . We identify this product explicitly. Zero occurs once at each level, producing x^m .

The positive value 2^j occurs at level $k = j$ through the rank-zero term, and at levels

$$j + 1 \leq k \leq \min(2j, m)$$

through a rank $2(k - j)$ quadratic sum. Its total multiplicity in the product is $1 + \min(j, m - j)$. The negative value -2^j occurs at the same latter levels but not at $k = j$, so its multiplicity is $\min(j, m - j)$. Consequently $\prod_k p_k = U_m$ and $U_m(T_Q) = 0$.

For any polynomial R , the action of $R(E)$ on $\mathbf{e}_\emptyset^\top T_Q^n \mathbf{e}_{S_{\text{full}}}$ is

$$\mathbf{e}_\emptyset^\top T_Q^n R(T_Q) \mathbf{e}_{S_{\text{full}}}.$$

Thus $U_m(E)$ annihilates every Fourier summand in (16), and hence the count itself. Since $U_m(E) = E^m P_m(E)$, the second assertion in (22) follows with the stated index range. Finally,

$$\deg P_m = \sum_{j=1}^m (2 \min(j, m - j) + 1), \quad \sum_{j=1}^m \min(j, m - j) = \lfloor m^2/4 \rfloor,$$

which yields (23). □

Corollary 4.3 (Form of every tail). *For $n \geq m$, there are polynomials $A_j, B_j \in \mathbb{Q}[n]$ such that*

$$N_{m,\mathcal{A}}(n) = \sum_{j=1}^m (A_j(n) 2^{jn} + B_j(n) (-2^j)^n), \quad (24)$$

with

$$\deg A_j \leq \min(j, m - j), \quad \deg B_j \leq \min(j, m - j) - 1.$$

A negative degree bound means the polynomial is zero. The generating function is rational. A candidate of the form (24) is certified by checking exactly $\deg P_m$ consecutive values beginning at $n = m$.

Proof. For $\lambda \neq 0$,

$$(E - \lambda)(\lambda^n p(n)) = \lambda^{n+1}(p(n+1) - p(n)).$$

Repeated application lowers the polynomial degree. The sequences $n^h \lambda^n$ for distinct nonzero λ and the prescribed degree ranges are linearly independent and give $\deg P_m$ solutions of the monic recurrence. Independence follows, for example, by applying the factors belonging to all roots except one and then taking finite differences at the remaining root. A recurrence of order $\deg P_m$ is uniquely determined by that many initial values, so these sequences span its solution space. The coefficients are rational because the corresponding linear system has rational entries and a unique solution. Their generating functions are rational; changing a finite prefix only adds a polynomial. The same uniqueness proves the certification claim. $\square$

4.2 The four-row specialization

The level spectra are particularly small:

Level k	Number of states	Permitted staying sums Λ_k
1	1	0, 2
2	3	0, -2, 2, 4
3	3	0, -4, 4, 8
4	1	0, -8, -4, 4, 8, 16

Multiplying the four level polynomials gives $U_4 = x^4 P_4$, with P_4 exactly as in (3). Therefore (4) is proved for the target count and, in fact, for every four-row Gram acceptance condition. The order 12 is a consequence of the combinatorial construction and the quadratic-sum lemma; it is not selected because twelve values happened to fit a formula.

5 The finite certificate and proof of the formula

5.1 Encoding the finite calculation

For reproducibility, the bit-mask implementation orders the ten Gram coordinates as

$$(11), (12), (13), (14), (22), (23), (24), (33), (34), (44). \quad (25)$$

The first coordinate is the least significant bit of the Gram mask. State bit $i - 1$ says that adjacent rows $i, i + 1$ have been resolved. A column mask uses its least significant bit for its top entry. These choices affect only the internal encoding, not the lexicographic convention for rows.

The acceptance test reconstructs each symmetric Gram matrix, reads each row from left to right, and checks strict decrease. It gives exactly 48 acceptable matrices. A useful small classification is

First row	Binary integer	Number of acceptable completions
1000	8	9
1100	12	14
1110	14	18
1111	15	7
Total		48

There are no other first rows. Indeed, decreasing rows imply a nonincreasing first column, symmetry identifies that column with the first row, and an all-zero first row cannot be greater than the second row. For each of the four possibilities, the six remaining upper-triangular bits give just 64 completions to test. The full list of accepted masks is included in the JSON certificate.

Finite certificate 5.1 (Definition-based integer counts). The dynamic program (8)–(10) gives the following values. The auxiliary column $z(n)$ counts the same increasing-row matrices with $MM^T = 0$; it is used only for the sharpness result in Section 6.3.

n	$a(n)$	$z(n)$
4	58	3
5	1 629	15
6	28 924	1 020
7	507 052	8 820
8	8 211 776	202 608
9	133 693 904	2 656 080
10	2 140 571 200	46 626 240
11	34 361 115 072	706 770 240
12	549 587 348 992	11 575 711 488
13	8 798 356 254 976	182 670 808 320
14	140 744 002 571 264	2 939 896 028 160
15	2 252 082 614 856 704	46 875 280 757 760

At every displayed index, exact rational evaluation of the right side of (2) equals $a(n)$. The same enumeration gives zero at $n = 0, 1, 2, 3$ for both columns.

The complete small program in Section B calculates the $a(n)$ column without importing any tabulated sequence values. It uses tuple states and tuple Gram coordinates. The separate program `verify.py` uses bit masks, also calculates $z(n)$, and performs the additional audits in Section 8. Thus the certificate is an explicit finite arithmetic claim, not an appeal to the authority of a pre-existing data table.

5.2 Why the twelve checks finish the proof

Let $F(n)$ denote the expression on the right side of (2), and put

$$Q(x) = (x - 16)(x - 8)^2(x + 8)(x - 4)^3(x + 4)^2(x - 2)^2. \quad (26)$$

The finite-difference identity in the proof of Theorem 4.3 shows directly that $Q(E)F = 0$ for all nonnegative indices. As $P_4(x) = (x + 2)Q(x)$, also $P_4(E)F = 0$.

Proof of Theorem 1.1. By Theorem 4.2, $P_4(E)a(n) = 0$ for $n \geq 4$. The preceding observation shows that the difference $h(n) = a(n) - F(n)$ satisfies that same order-12 monic recurrence on this range. Theorem 5.1 gives $h(4) = h(5) = \dots = h(15) = 0$.

Write $P_4(x) = x^{12} + \sum_{j=0}^{11} p_j x^j$. The recurrence at $n = 4$ gives $h(16) = -\sum_{j=0}^{11} p_j h(4+j) = 0$. Inductively, if h vanishes through $n + 11$, the recurrence at n gives $h(n + 12) = 0$. Hence $h(n) = 0$ for every $n \geq 4$. The initial zeros come from the finite certificate, or directly from the same exact enumeration for these four small lengths. $\square$

Remark 5.2 (The transient is real). The lower limit $n \geq 4$ must not be dropped. The extrapolated expression gives

$$F(0) = \frac{103}{4096}, \quad F(1) = -\frac{71}{1024}, \quad F(2) = -\frac{1}{64}, \quad F(3) = -\frac{1}{16},$$

whereas the counting sequence is zero at all four indices. The zero root in the universal annihilator accounts for such finite transients.

6 Generating functions, minimal recurrences, and sharpness

6.1 The exact rational generating function

Define

$$D(z) = (1 - 16z)(1 - 8z)^2(1 + 8z)(1 - 4z)^3(1 + 4z)^2(1 - 2z)^2 \quad (27)$$

and

$$\begin{aligned} B(z) = & 58 - 227z - 9052z^2 + 68976z^3 - 16960z^4 \\ & - 1025408z^5 + 3055104z^6 - 3796992z^7 \\ & + 2834432z^8 - 2490368z^9 + 2097152z^{10}. \end{aligned} \quad (28)$$

Corollary 6.1 (Rational generating function). *As an identity of formal power series,*

$$\sum_{n \geq 0} a(n)z^n = z^4 \frac{B(z)}{D(z)}. \quad (29)$$

Equivalently, $\sum_{k \geq 0} a(k+4)z^k = B(z)/D(z)$.

Proof. By Theorem 1.1, the shifted sequence $b(k) = a(k+4)$ is annihilated by $Q(E)$ for every $k \geq 0$. The reciprocal characteristic polynomial is $D(z) = z^{11}Q(1/z)$. If $D(z) = \sum_{j=0}^{11} d_j z^j$, the coefficients of $D(z) \sum_{k \geq 0} b(k)z^k$ vanish from degree 11 onward. For $0 \leq k \leq 10$ they are the finite convolutions

$$\sum_{j=0}^k d_j a(k-j+4).$$

Using Theorem 5.1 gives exactly (28). The four initial zero terms then supply the factor z^4 . □

For direct implementation the characteristic polynomial is

$$\begin{aligned} Q(x) = & x^{11} - 32x^{10} + 244x^9 + 1552x^8 - 27392x^7 \\ & + 70144x^6 + 484352x^5 - 2781184x^4 + 1212416x^3 \\ & + 20185088x^2 - 48234496x + 33554432. \end{aligned} \quad (30)$$

Thus $\sum_{j=0}^{11} q_j a(n+j) = 0$ for $n \geq 4$, where q_j is the coefficient of x^j in (30). The range is important: this order-11 recurrence fails at each of $n = 0, 1, 2, 3$.

6.2 Minimality for the selected sequence

Proposition 6.2. *The minimal monic constant-coefficient annihilator of the eventual tail of $a(n)$ is $Q(x)$, of degree 11. If a recurrence is required to hold at every $n \geq 0$, its minimal monic annihilator is $x^4Q(x)$, of degree 15.*

Proof. In (2), the six distinct bases and the exact polynomial degrees are

λ	16	8	-8	4	-4	2
$\deg p_\lambda$	0	1	0	2	1	1.

Each leading coefficient is nonzero. The independence and finite-difference argument from Theorem 4.3 imply that a polynomial annihilates $p_\lambda(n)\lambda^n$ only if it contains $(x - \lambda)$ to at least $1 + \deg p_\lambda$. Distinct roots cannot cancel one another's components. Therefore the minimal eventual annihilator is precisely (26).

For the global assertion, write $a = F + \delta$, where δ is supported on 0, 1, 2, 3 and $\delta(3) = 1/16 \neq 0$. The forward shift on sequences supported in this range is nilpotent, and δ has nilpotency index exactly four. Any global annihilator must be divisible by Q , because its action on the exponential-polynomial tail must vanish. Since $Q(0) \neq 0$, the operator $Q(E)$ is invertible on this finite-support space and preserves that nilpotency index. Consequently a further factor x^4 is necessary and sufficient to kill the transient. This proves the second assertion. $\square$

There is no conflict between an eventual recurrence order of 11 here and the order-10, degree-one polynomial-coefficient guess reported in the 2023 paper: constant-coefficient order and polynomial-coefficient order are different notions. The minimality assertion above concerns constant coefficients only.

6.3 The four-row universal bound is sharp

It is natural to ask whether the factor $x + 2$ in P_4 was only an artifact of the argument: it disappears from the answer for $\mathcal{A}_*$. It is not an artifact for arbitrary acceptance sets.

Proposition 6.3 (Sharp universal order for four rows). *Among all polynomials that eventually annihilate $N_{4,\mathcal{A}}$ for every $\mathcal{A} \subseteq \mathcal{G}_4$, the minimal monic one is P_4 . Its degree is 12. If annihilation must hold from $n = 0$, the minimal common monic polynomial is x^4P_4 , of degree 16.*

Proof. Theorem 4.2 supplies both upper bounds. A common eventual annihilator must be divisible by Q , by Theorem 6.2. To see the missing factor, use the target $\mathcal{A} = \{0\}$ and write $z(n) = N_{4,\{0\}}(n)$.

The universal theorem says that on $n \geq 4$, the only component of $z(n)$ not annihilated by $Q(E)$ is a constant multiple $c(-2)^n$, because the multiplicity of -2 in P_4 is one. The exact counts in Theorem 5.1, substituted into (30), give

$$Q(E)z(4) = 136857600, \quad Q(-2) = 149299200.$$

It follows that

$$c = \frac{136857600}{(-2)^4 \cdot 149299200} = \frac{11}{192} \neq 0. \quad (31)$$

Hence $x + 2$ is also necessary. Its product with Q is P_4 , proving eventual sharpness.

For a common annihilator valid from $n = 0$, the selected sequence a already forces x^4Q by Theorem 6.2; the zero-Gram target forces $x + 2$. Their least common multiple is x^4P_4 , completing the proof. $\square$

The degrees 11, 12, 15, and 16 therefore answer four different questions, not four inconsistent descriptions of the same recurrence.

7 Equidistribution and exact asymptotic information

7.1 A general uniformity theorem

Theorem 7.1 (Ordered Gram equidistribution). *For each fixed $m \geq 2$ and every $\mathcal{A} \subseteq \mathcal{G}_m$,*

$$N_{m,\mathcal{A}}(n) = \frac{|\mathcal{A}|}{2^{m(m+1)/2}} \binom{2^n}{m} + O_m(n 2^{(m-1)n}) \quad (n \rightarrow \infty). \quad (32)$$

In particular, if an increasing-row matrix is chosen uniformly, its Gram matrix tends to the uniform distribution on all symmetric $m \times m$ binary matrices, and

$$\Pr(MM^\top \in \mathcal{A}) = \frac{|\mathcal{A}|}{2^{m(m+1)/2}} + O_m(n 2^{-n}). \quad (33)$$

The implied constants may be chosen uniformly over $\mathcal{A}$ for fixed m .

Proof. The zero character in (16) contributes exactly $2^{-d} |\mathcal{A}| \binom{2^n}{m}$, by (11). Consider a nonzero character Q . At every level $k < m$, a diagonal sum has absolute value at most $2^k \leq 2^{m-1}$. At level m , the function q_Q in (14) is nonzero: its values on e_i recover Q_{ii} , and its values on $e_i + e_j$, together with those single-coordinate values, recover Q_{ij} . Thus q_Q cannot be identically zero unless $Q = 0$.

If the polarization of this nonzero q_Q has rank zero, its Walsh sum is zero. Otherwise its rank is at least two, and Theorem 3.1 bounds the sum by 2^{m-1} in absolute value. Hence no nonzero character has the eigenvalue 2^m .

There is also a sharp enough multiplicity bound. The value $+2^{m-1}$ can occur only at levels $m-1$ and m , so its polynomial prefactor has degree at most one. The value -2^{m-1} can occur only at level m , so its prefactor is constant. Every other nonzero eigenvalue has smaller absolute value and a bounded polynomial prefactor, and a zero-eigenvalue component has finite support. Each nonzero Fourier summand is consequently $O_m(n 2^{(m-1)n})$. There are finitely many characters, with $|\widehat{\mathbf{1}_{\mathcal{A}}}(Q)| \leq 2^d$, so summing preserves this estimate uniformly over the target set. Division by $\binom{2^n}{m} \sim 2^{mn}/m!$ proves (33). $\square$

The sorting of rows is relevant here: conditional Gram uniformity should not simply be assumed from an unconstrained random-matrix model. The proof shows why sorting affects only exponentially smaller terms.

7.2 Leading terms for A181280

Because $|\mathcal{A}_*| = 48$ and $2^d = 1024$, Theorem 7.1 predicts the limiting conditional probability $48/1024 = 3/64$. The exact formula gives more:

$$a(n) = \frac{16^n}{512} + \left(\frac{n}{512} - \frac{3473}{147456} - \frac{113(-1)^n}{49152} \right) 8^n + O(n^2 4^n), \quad (34)$$

$$\frac{a(n)}{\binom{2^n}{4}} = \frac{3}{64} + \left(\frac{3n}{64} - \frac{1745}{6144} - \frac{113(-1)^n}{2048} \right) 2^{-n} + O(n^2 4^{-n}). \quad (35)$$

Thus $a(n) \sim 16^n/512$. Among *all* binary $4 \times n$ matrices, the probability of satisfying both ordering conditions tends to $1/512$; among those with already increasing rows it tends to $3/64$. These are different sample spaces and should not be conflated.

The original count has a finite exact exponential-polynomial expansion: (2) terminates after absolute exponential scales $16^n, 8^n, 4^n$ and 2^n , with parity effects at the middle two scales. There is no unproved asymptotic remainder hidden in that identity.

7.3 All coefficients of the conditional-probability expansion

A compact exact rational form gives an infinite expansion for the conditional probability as well. Put $x = 2^{-n}$ and $\varepsilon = (-1)^n$, and define

$$\begin{aligned} v_0 &= \frac{3}{64}, \\ v_1(n, \varepsilon) &= \frac{288n - 3473}{6144} - \frac{113\varepsilon}{2048}, \\ v_2(n, \varepsilon) &= \frac{6n^2 - 219n + 820 - \varepsilon(13n - 164)}{256}, \\ v_3(n, \varepsilon) &= -\frac{3n + 32}{12}. \end{aligned}$$

Since $\binom{2^n}{4} = 2^{4n}(1-x)(1-2x)(1-3x)/24$, the exact identity is

$$\frac{a(n)}{\binom{2^n}{4}} = \frac{v_0 + v_1x + v_2x^2 + v_3x^3}{(1-x)(1-2x)(1-3x)} \quad (n \geq 4). \quad (36)$$

Let $h_t = 0$ for $t < 0$ and, for $t \geq 0$, let

$$h_t = \frac{1 - 2^{t+3} + 3^{t+2}}{2}. \quad (37)$$

Partial fractions give

$$\frac{1}{(1-x)(1-2x)(1-3x)} = \frac{1/2}{1-x} - \frac{4}{1-2x} + \frac{9/2}{1-3x} = \sum_{t \geq 0} h_t x^t.$$

It follows that the coefficient polynomials in the full expansion are

$$C_k(n, \varepsilon) = \sum_{j=0}^3 v_j(n, \varepsilon) h_{k-j}, \quad \frac{a(n)}{\binom{2^n}{4}} = \sum_{k \geq 0} C_k(n, (-1)^n) 2^{-kn}. \quad (38)$$

For each fixed $n \geq 4$ the latter series converges absolutely. For a fixed truncation order K , its tail is $O_K(n^2 2^{-(K+1)n})$ as $n \rightarrow \infty$. Indeed $h_t = O(3^t)$, all v_j are $O(n^2)$, and $3 \cdot 2^{-n} < 1$ on this range. This supplies every coefficient without further recurrence guessing.

8 Computation, reproducibility, and audit boundaries

8.1 What was executed

The supplied standard-library program `verify.py` was executed using Python 3.13.5. Its recorded run completed in approximately 4.15 seconds on the working machine; timing is an observation, not a performance guarantee. The output is saved in `results/verification.txt`.

The executed checks comprise exact dynamic programming through $n = 100$; comparison with (2) throughout $4 \leq n \leq 100$; the unrestricted count (11); all 1024 weighted 8×8 identities $U_4(T_Q) = 0$; Fourier inversion through $n = 15$; and the displayed generating function and recurrence coefficients. Every nonzero-character matrix also passes the spectral-radius bound eight directly on its diagonal.

A separate brute-force enumeration chooses all four-element subsets of binary row words for $0 \leq n \leq 6$ and computes every Gram row directly using parity of bit intersections. At $n = 6$ this checks $\binom{64}{4} = 635376$

row sets. It uses neither order-state transitions nor upper-triangular Gram masks. Both the A181280 count and the zero-Gram count agree with the dynamic program. For $m = 1, 2, 3$, the program additionally checks the universal recurrence through $n = 20$ for *every* individual Gram target.

Finally, `minimal_certificate.py`, reproduced in Section B, was executed independently. It represents states and Gram entries by tuples rather than masks and reproduces all twelve decisive values. The scripts use exact integers and rational arithmetic only. They perform no network requests and read no sequence data as input.

8.2 What makes this a finite proof rather than a large test

The mathematical implication is

$$\text{definition} \implies U_4(E)a = 0 \implies P_4(E)a = 0 \text{ on } n \geq 4,$$

followed by uniqueness of solutions of a monic order-12 recurrence and a finite arithmetic certificate. The extra computations through $n = 100$ are useful cross-checks, but the all- n conclusion does not depend on assuming that a pattern observed through 100 continues.

The computer-assisted dependency is transparent: the arithmetic in Theorem 5.1 and in the small acceptance enumeration must be correct. The transition invariant, Fourier identity, quadratic-sum argument, recurrence bound, and finite-to-infinite induction are conventional proofs given in full. The two state encodings and the independent brute force reduce implementation risk; they do not replace mathematical review. No Lean formalization or external peer review is claimed.

8.3 Complexity and practical use

The direct state space has 2^{d+m-1} entries and at most 2^m column choices per state. A straightforward implementation through length N uses

$$O(N2^{d+2m-1})$$

integer additions and two layers of $O(2^{d+m-1})$ integer registers, apart from optional stored history. Counts have $O(mN)$ bits. Thus for fixed m the arithmetic-operation count is linear in N , while a naive bit-operation bound for the additions is quadratic in N . The full verifier retains endpoint Gram distributions at all lengths for its audits, so its stored history uses $O(N2^d)$ integer registers in addition to the working layers.

For the selected sequence, once the formula is established, either (2) or the order-11 recurrence is preferable for computing very large indices. An isolated value can be evaluated with integer exponentiation and a fixed amount of rational arithmetic. This does not imply polylogarithmic bit complexity: the answer itself has $\Theta(n)$ bits.

The reusable enumerator explicitly limits its implementation to $1 \leq m \leq 5$ to avoid accidental exponential resource consumption. This is a software guard, not a limitation of Theorem 4.2. The present exhaustive Fourier audit was performed for $m = 4$, not for all m .

9 Conclusions and remaining research directions

The explicit A181280 identity has a complete proof here, with a small, reproducible computer-assisted arithmetic component. The decisive step is the universal spectral bound, not a more aggressive fit of the

existing data. The factor $(x + 2)$ vanishes for the decreasing-Gram condition but is necessary for other targets, and the four exceptional initial indices are explained by a genuine nilpotent transient. The rational generating function, minimal recurrences, and asymptotic probability all follow from the same calculation.

For general m , the theorem gives a uniform quadratic-size recurrence bound despite an exponentially large finite-state realization. What remains to be determined by further work is when that universal bound is sharp for other row counts, which target conditions cancel particular spectral components, and whether explicit structural formulas can replace the remaining finite arithmetic for broad families of target sets. These are directions suggested by the present results, not additional open problems asserted to have been settled here.

The status distinction remains important. C-finiteness was already proved in the retrieved 2025 slides, and the explicit formula was already conjectured in 2023. The present package supplies its proof and the additional structural results developed above. Establishing priority would require a fuller literature and author check than the available-source audit carried out here.

A The recurrence certificate in machine-readable form

The file `results/certificate.json` records the twelve initial values, the 48 accepted Gram masks in the coordinate order (25), all recurrence and generating-function coefficient arrays, the independent brute-force results, and the Fourier audit summary. Polynomial arrays are always listed in increasing powers.

For a convenient manual comparison, the universal eventual polynomial is

$$\begin{aligned} P_4(x) = & x^{12} - 30x^{11} + 180x^{10} + 2040x^9 - 24288x^8 + 15360x^7 \\ & + 624640x^6 - 1812480x^5 - 4349952x^4 + 22609920x^3 \\ & - 7864320x^2 - 62914560x + 67108864. \end{aligned}$$

The four nonzero early residuals of the smaller recurrence are

$$(Q(E)a(0), Q(E)a(1), Q(E)a(2), Q(E)a(3)) = (-3796992, 2834432, -2490368, 2097152).$$

They provide an especially simple check that an implementation has not silently shifted the recurrence's valid range.

The observed level spectra in the exhaustive Fourier audit were exactly $\Lambda_1, \Lambda_2, \Lambda_3, \Lambda_4$ as listed in Section 4. This observation is supplementary: the proof only needs containment in those sets, which follows analytically from Theorem 3.1.

B A complete small arithmetic verifier

The following is the complete file `minimal_certificate.py`. Its ordinary list comparisons implement left-to-right lexicographic order. It proves the finite arithmetic portion of Theorem 1.1; the universal recurrence theorem supplies the all- n implication.

```
#!/usr/bin/env python3
"""Exact twelve-value certificate. Python 3.10+, standard library only.
The article proves the all-n implication; this code uses tuple states.
"""
from collections import defaultdict
from fractions import Fraction as F
from itertools import product

pairs = [(i, j) for i in range(4) for j in range(i, 4)]
states = list(product((False, True), repeat=3))
columns = list(product((0, 1), repeat=4))
full = (True, True, True)

def accepted(g):
    matrix = [[0] * 4 for _ in range(4)]
    for (i, j), bit in zip(pairs, g):
        matrix[i][j] = matrix[j][i] = bit
    return matrix[0] > matrix[1] > matrix[2] > matrix[3]

def formula(n):
    return (F(16**n, 512)
            + F((288*n - 3473)*8**n, 147456)
            - F(113*(-8)**n, 49152)
            + F((6*n*n - 219*n + 820)*4**n, 6144)
            - F((13*n - 164)*(-4)**n, 6144)
            - F((3*n + 32)*2**n, 288))

edges = {}
for state in states:
    edges[state] = []
    for v in columns:
        if any(not state[i] and v[i] > v[i+1] for i in range(3)):
            continue
        target = tuple(state[i] or v[i] < v[i+1] for i in range(3))
        delta = tuple(v[i] * v[j] for i, j in pairs)
        edges[state].append((target, delta))

accepting = {g for g in product((0, 1), repeat=10) if accepted(g)}
if len(accepting) != 48:
    raise AssertionError('Acceptance predicate is incorrect')

tab = {((False, False, False), (0,) * 10): 1}
counts = []
for n in range(16):
    counts.append(sum(value for (state, gram), value in tab.items()
                      if state == full and gram in accepting))
    if n == 15:
        break
    nxt = defaultdict(int)
    for (state, gram), value in tab.items():
        for target, delta in edges[state]:
            new_gram = tuple(x ^ y for x, y in zip(gram, delta))
            nxt[target, new_gram] += value
    tab = nxt

if counts[:4] != [0, 0, 0, 0]:
    raise AssertionError('Initial prefix differs')
for n in range(4, 16):
    if counts[n] != formula(n):
        raise AssertionError(f'The certificate fails at n={n}')
    print(n, counts[n], 'OK')
print('All twelve exact certificate values match.')
```

C Package contents and reproduction

The main article is `article.tex`, compiled to `article.pdf`. Both Python programs need only Python 3.10 or later. The `results/` directory contains the generated JSON certificate, CSV counts through $n = 100$, and executed audit output. `STATUS.md` records the source-status distinctions; `README.md` contains usage instructions; `build.sh` rebuilds this PDF with pdfLaTeX.

From the package root, the principal commands are

```
python3 minimal_certificate.py
python3 verify.py --out results
sh build.sh
```

The article does not depend on a symbolic-algebra package. The TeX source uses standard TeX Live packages, including `newtx`, `amsmath`, `tcolorbox`, `listings`, and `hyperref`. No font files or third-party source documents are included in the archive.

References

- [1] R. H. Hardin, *A181280: Number of $4 \times n$ binary matrices with increasing rows and mod-2 Gram matrix with decreasing rows*, The On-Line Encyclopedia of Integer Sequences. Entry introduced 10 October 2010; formula attributed to M. Kauers and C. Koutschan on 2 March 2023. <https://oeis.org/A181280>. Retrieved 19 September 2026.
- [2] M. Kauers and C. Koutschan, *Some D-Finite and Some Possibly D-Finite Sequences in the OEIS*, Journal of Integer Sequences **26** (2023), Article 23.4.5. See Section 6.5, Conjecture 20. <https://cs.uwaterloo.ca/journals/JIS/VOL26/Koutschan/kout4.html>. Author version: arXiv:2303.02793v2, 24 April 2023, <https://arxiv.org/html/2303.02793v2>. Retrieved 19 September 2026.
- [3] M. Kauers, *Guessing with little data*, Rutgers Experimental Mathematics Seminar, 11 December 2025, lecture slides. Formula on slide 30 (PDF page 104); general C-finiteness theorem on slide 39 (PDF page 137); overview check marks on slides 24 and 40. <https://sites.math.rutgers.edu/~zeilberg/expmath/kauers25.pdf>. Retrieved 19 September 2026.
- [4] Rutgers Experimental Mathematics Seminar, *Archive of Speakers and Talks—2025*, entry for 11 December 2025. <https://sites.math.rutgers.edu/~zeilberg/expmath/archive25.html>. Retrieved 19 September 2026.